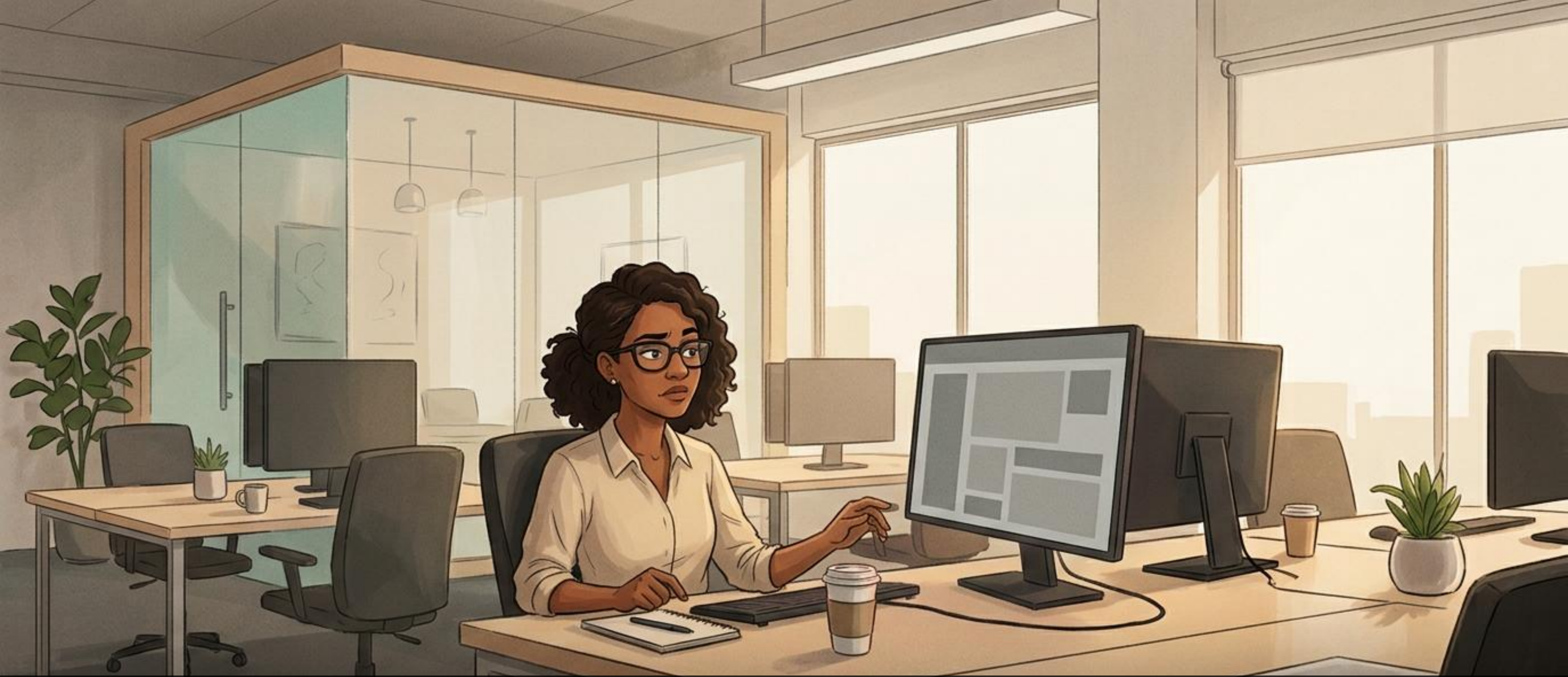
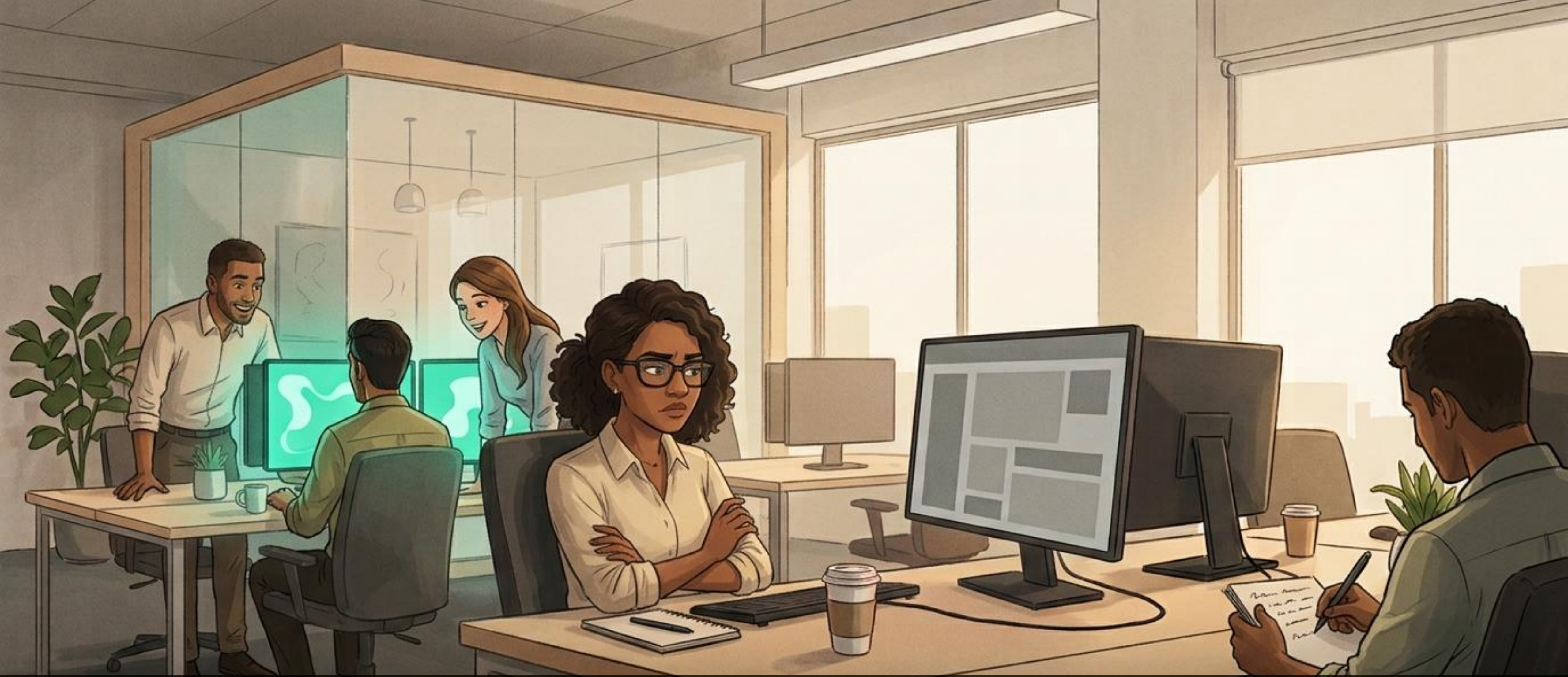
Rohan built a skill his team loved.

COLD OPEN · SCENE 01



For Rohan it was magic. For Priya, nothing.

COLD OPEN · SCENE 02



Half the team quietly gave up on it.

COLD OPEN · SCENE 03

The skill was fine — it just never fired.

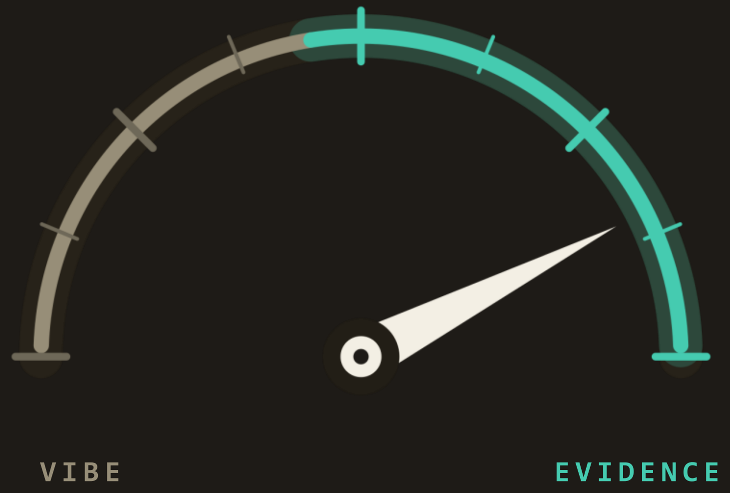
COLD OPEN · SCENE 04

EVALUATING AGENT SKILLS · A PRACTITIONER'S PLAYBOOK

Stop Vibe-Shipping Your Skill

Is it **sound**? Does it **fire**? Does it **help**?

A practitioner's guide to evaluating the agent skills you ship — from vibe-checks to evidence.



READING: → EVIDENCE

Tezan Sahu

Applied Scientist · M365 Copilot Extensibility, Microsoft · Common Ground 2026

HI, I'M YOUR HOST

Tezan Sahu

Applied Scientist 2 · M365 Copilot Extensibility, Microsoft

IIT Bombay — Institute Rank 2 || **8** US patents || **Bestselling Author** of **Beyond Code**

Writes **Low-Pass Filter** — a FREE signal-over-hype AI Newsletter

OFF THE CLOCK

- **Author**
- **Trained vocalist**
- **Racket-sport nerd**
- **Artist · 3 exhibitions**

THE PROBLEM, NAMED

Welcome to Skill Hell.

*Framework hell → tutorial hell → **skill hell**. Same root cause: abundance, no evaluation.*

> *"You have all these skills, freely available to download — but you don't really know how the pieces work together, **which ones actually work**, or why the agent keeps ignoring the ones you're sure should fire."*
>
> – MATT POCOCK, ON COINING "SKILL HELL"

THE MENTAL MODEL

- **The LLM is the operating system**
- **Skills are the apps you install**
- **...and there is no App Review**

~4k	36.8%	13.4%	~1 in 5
skills on one marketplace	flawed on inspection	carried critical issues	curated skills make it WORSE

THE BAR TO PUBLISH TODAY: A SKILL.md and a one-week-old account. No signing. No review.

Evaluating a skill is really three questions.

Two of them are new here — and they only count if you actually run the skill.

01

Is it sound?

INTRINSIC · STATIC

Well-built, readable, safe. Reads the skill; never runs it.

This is validation.

02

Does it fire?

TRIGGERING · RUNTIME

The right skill wakes up on the right prompt — and stays quiet otherwise.

This is evaluation.

03

Does it help?

BEHAVIOUR · RUNTIME

When it runs, the output actually beats the agent without it.

This is evaluation.

VALIDATION — reads the skill

EVALUATION — runs the skill · the part almost nobody does

Your skill passed lint. Did it pass the point? Validation is table stakes. Evaluation is the two questions that follow.

Is it sound? Read the skill before you run it.

Cheap, automatic, static — the gate you keep, but never mistake for proof.

Structural validity

Folder parses · frontmatter valid · links resolve · no secrets leaking.

Quality & safety

Readable, complete, well-authored · free of dangerous code patterns.

One command, one scan.
A static evaluator scores it and a security scanner reads the code.
No LLM required.

SAME JOB — ONE CLEARS, ONE DOESN'T

Bundles a script
hardcoded key · shell-out · eval()
✗ DO NOT INSTALL

Prose-only, tight scope
no scripts · least privilege
✓ SAFE TO RUN

Necessary — not sufficient. Sound means safe to run — nothing yet about whether it fires right, or helps.

Does it fire? A perfect skill that never wakes up is worthless.

The model decides when to load a skill — and it gets that decision wrong in both directions.

MISFIRE
Wakes on the wrong prompt — derails the answer.

NO-FIRE
Stays asleep when it was perfect for the job.

> *"The model may choose not to follow a context pointer even when it's perfect for the task."*
>
> — which is exactly why you have to eval whether it fires.

THE TEST — e.g. an "ACTION-ITEMS" SKILL

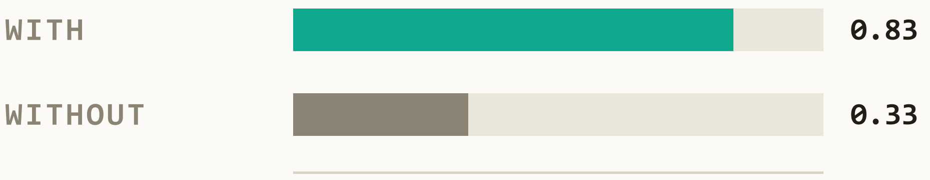
SHOULD FIRE	"transcript → action items"	fired	✓
SHOULD FIRE	"who owns what here?"	fired	✓
STAY QUIET	"summarise this call"	quiet	✓
STAY QUIET	"draft a follow-up email"	quiet	✓
STAY QUIET	"what time did we start?"	quiet	✓

PRECISION 1.00 · RECALL 1.00 · 3 near-misses caught

Does it help? Run it twice — with the skill, and without.

The delta is the whole verdict — and it's the one check almost nobody runs.

EXHIBIT — WITH vs WITHOUT



Δ +0.50 pass-rate · +13s · +1.7k tok

Same prompts, run twice — once with the skill loaded, once without.

READING THE DELTA

- ✓ **Big lift, small cost** → ship it.
- ✗ **Tiny lift, big cost** → don't.
- – **Passes with AND without** → dead weight.

The delta is the only number that says a skill earns its place.
~1 in 5 curated skills are net-negative — eyeballing never catches it.

Author the eval sets: what should fire, and what “good” looks like.

Two runtime checks, two tiny datasets — both mined from real scenarios, not invented happy paths.

TRIGGERING QUERYSET

for “does it fire?”

SHOULD FIRE

prompts that must wake the skill

NEAR-MISS

share the words, wrong job — must stay quiet

~3 should-fire + ~3 near-miss → score precision & recall

QUALITY EVALSET

for “does it help?”

Prompt	a realistic user message
Expected	plain-English “what good looks like”
Assertions	one checkable fact each — write AFTER run 1

~3 prompts → run each WITH and WITHOUT

Keep it demo-sized — ~6 queries, ~3 prompts, plus one edge case. Small enough to read every row, real enough to matter.

Grade the outcome — not the path.

The agent will reach a good answer a way you never scripted. Grade what it produced, not how.

THE PRINCIPLE

An agent reaches a correct answer by a route you didn't predict. Grade the route, and every good run fails your eval.

✗ `assert: called tool A, then B, then C`

brittle — breaks on any new-but-valid path

✓ `assert: output cites every required figure`

robust — grades the result, not the journey

GRADE WITH THE CHEAPEST METHOD THAT WORKS

1. **Deterministic** — valid JSON · counts · file exists
2. **Script checks** — mechanical, reusable, free
3. **LLM-as-judge** — boolean + one line of evidence
4. **Human** — for what assertions can't catch

Run each case 3–5×.
Same prompt, different output — read the distribution, not one lucky pass.

THE GOOD NEWS

You don't build the harness. It already exists.

Two open-source skills run the entire loop — including the one you already author skills with.

01
Generate
queryset + evalset

→

02
Run
with vs without

→

03
Grade
passed + evidence

→

04
Benchmark
pass-rate & cost

→

05
Review
an HTML scorecard

skill-creator

ANTHROPIC · OPEN SOURCE

The same skill you author skills with also runs the full eval loop.

evaluate-skill-with-copilot-sdk

OPEN SOURCE · BUILT ON SKILL-CREATOR

Drives isolated Copilot SDK sessions for BOTH triggering (Path A) and quality (Path B).

Let's put a real skill on the bench.

One scenario — a competitive research brief — run through all three questions, end to end.

01 · Is it sound?

Two skills, same job. The scanner rejects the foil; the hero clears.

SKILL-EVALUATOR

→

02 · Does it fire?

Should-fire vs near-miss prompts → precision & recall on the hero.

COPILOT-SDK · PATH A

→

03 · Does it help?

With vs without on frozen sources → a real, positive delta.

COPILOT-SDK · PATH B

Safety net: every step has a pre-baked exhibit — the intrinsic reports, the triggering dashboard, and the with/without scorecard — so a flaky network never stalls the story.

Next: skills that evolve themselves.

Automatic optimization — the skill rewrites itself from its own eval signal.

THE OPTIMIZATION LOOP

Measure
score on its eval set

→

Propose
model rewrites weak spots

→

Re-measure
run the delta again

→

Keep
accept only edits that win

↻ repeat until the delta stops improving

WHERE THE RESEARCH IS GOING

Self-discovery · **SEAgent**
agents mine their own skills

RL in the loop · **SAGE**
skills become the reward signal

Self-refinement · **Memento-Skills**
an agent that designs its own

Every one has an **evaluation loop at its core**.

WHAT TO TAKE BACK

Three moves. Starting Monday.

01

Stop vibe-shipping

Treat skills as infrastructure, not prose. “It looked fine” is not a test you can build a system on.

02

Run the runtime checks

Sound is table stakes. The proof: does it fire, and does it help. Measure the delta.

03

Let the harness do it

skill-creator and **evaluate-skill-with-copilot-sdk** run the whole loop today — and self-evolving skills are next. Adopt the shape now.

THE VERDICT

Anyone can make a skill.

From today, you can prove yours works — before you ship it.

Stop shipping on vibes.

Start measuring.

Thank you.

LinkedIn

in/tezan-sahu

Low-Pass Filter

the signal-over-hype newsletter

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